Future ONCOLOGY

Clinical Trial Protocol

SURPASS-ET: phase III study of ropeginterferon alfa-2b versus anagrelide as second-line therapy in essential thrombocythemia

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Patients diagnosed with high-risk essential thrombocythemia (ET) have limited treatment options to reduce the risk of thrombosis and lessen the progression of the disease by targeting the molecular source. Hydroxyurea is the recommended treatment, but many patients experience resistance or intolerance. Anagrelide is an approved second-line option for ET, but concerns of a higher frequency of disease transformation may affect its role as a suitable long-term option. Interferons have been evaluated in myeloproliferative neoplasms for over 30 years, but early formulations had safety and tolerability issues. SURPASS-ET (NCT04285086) is a phase III, open-label, multicenter, global, randomized, active-controlled trial that will evaluate the safety, efficacy, tolerability and pharmacokinetics of ropeginterferon alfa-2b compared with anagrelide as second-line therapy in high-risk ET.

Plain language summary: Essential thrombocythemia (ET) is a condition characterized by having more platelets than normal. The high number of platelets increases the risk of a life-threatening blood clot and/or bleeding. Patients with ET and at a high risk for these events are usually treated first with hydroxyurea (HU), but some patients do not respond properly or may develop significant side effects. Anagrelide is an approved medication used in patients who do not respond to HU. Ropeginterferon alfa-2b is a disease-specific, long-acting interferon with a good safety profile approved in polycythemia vera, another type of myeloproliferative neoplasm. The SURPASS-ET clinical trial will evaluate the safety, efficacy, tolerability and pharmacokinetics of ropeginterferon alfa-2b compared with anagrelide in patients with ET who are resistant or cannot tolerate HU.

Trial Registration Number: NCT04285086 (ClinicalTrials.gov)

Tweetable abstract: New #OpenAccess phase III #clinical trial protocol comparing ropeginterferon alfa-2b to anagrelide for second-line treatment in ET and includes the history of #interferon use in myeloproliferative neoplasms.

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Keywords: anagrelide • essential thrombocythemia • hydroxyurea • interferon • myeloproliferative neoplasm • P1101 • pegylated interferon • ropeginterferon alfa-2b • SURPASS-ET

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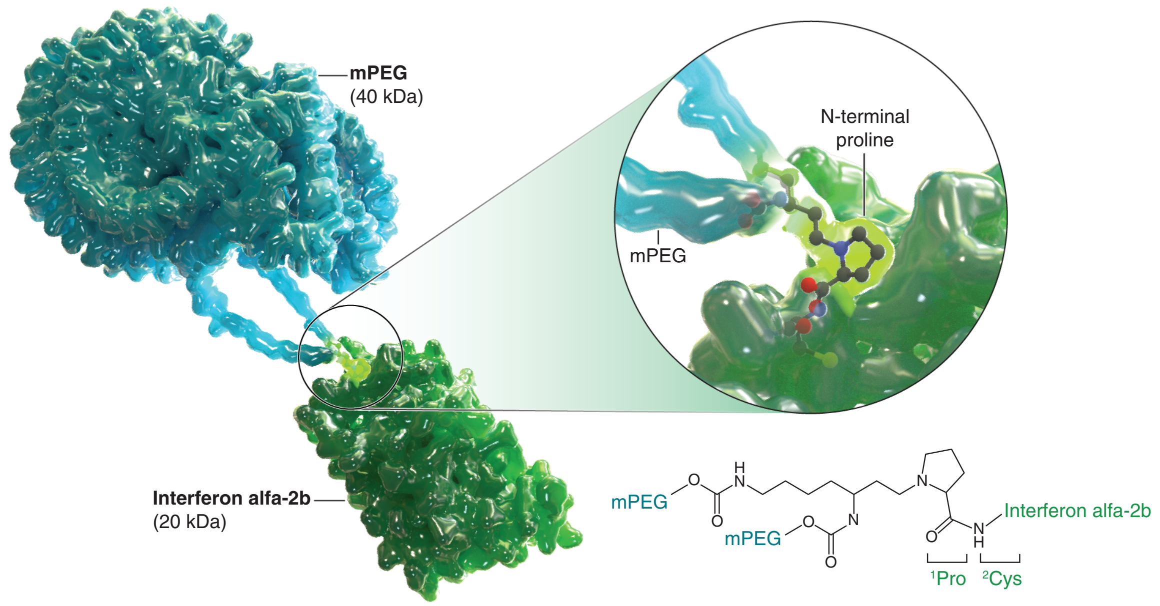

Figure 1. Molecular structure of ropeginterferon alfa-2b-njft.

Essential thrombocythemia (ET) is a subtype of chronic myeloproliferative neoplasm (MPN) characterized by thrombocytosis and is associated with a high risk of thrombosis and hemorrhage, and progression to secondary disease transformations [1,2]. Ideal therapeutic approaches should achieve adequate control of platelets (and white blood cells [WBC] in patients with leukocytosis), control ET-related symptom burden and splenomegaly, reduce the risk of thrombo-hemorrhagic complications (either first or recurrent) and prevent progression to post-ET myelofibrosis (PET-MF) or secondary acute myeloid leukemia (AML) [2–5]. Low-dose aspirin with hydroxyurea (HU) is currently recommended as first-line therapy in high-risk patients [2,6]. Studies with HU, a chemotherapeutic nondisease-specific cytoreductive, show approximately 20–25% of ET patients become HU intolerant or resistant and their disease is inadequately controlled [2,6–9]. Additionally, patients with resistance appear to be at an increased risk of disease transformation and reduced overall survival [4,5]. Real world evidence suggests up to 73% of patients discontinue HU due to adverse events and/or lack of efficacy [10]. Indeed, there is a paucity of prospective clinical data to guide the management of ET patients who are HU resistant or intolerant. The current alternative treatment options to HU, in other words, poly-pegylated IFN-α 2a which is used off-label and anagrelide (ANA), are associated with increased safety and tolerability concerns [11–13]. Furthermore, one study that evaluated over 800 high-risk ET patients treated with ANA and low-dose aspirin were observed to have higher incidence rates of thrombosis and myelofibrosis transformation compared with those treated with HU and low-dose aspirin after a median follow-up of 39 months [12].

Ropeginterferon alfa-2b

Ropeginterferon alfa-2b is a novel, site-specific, monopegylated, stable IFN-α analog with a structure that is seen in Figure 1 [14–17]. This unique single isoform differentiates ropeginterferon alfa-2b from earlier generation polypegylated IFN which utilized random pegylation methods and therefore, contain many isoforms with each polyethylene glycol (PEG) conjugate having its own activity and stability properties [14,17,18]. These structural characteristics gives ropeginterferon alfa-2b beneficial pharmacokinetic and pharmacodynamic properties allowing for less frequent dosing with a favorable safety and tolerability profile [14,17].

Ropeginterferon alfa-2b-njft (BESREMi®) is currently approved for the treatment of adults with polycythemia vera (PV) in the USA based on efficacy data from PEGINVERA and pooled safety data from the PROUD-PV/CONTINUATION-PV and PEGINVERA studies [19–21]. PEGINVERA was a phase II, prospective, multi-

Table 1. Listing of clinical studies evaluating the role of interferon in the treatment and management of myeloproliferative neoplasms.

First study (year)	ET patients (n)	Response rate, %	Discontinuation, n (%)	Type of IFN
Giles (1988)	18	100	0	2a and 2b
Bellucci (1988)	12	NA	4 (33)	2a
Gugliotta (1989)	10	100	NA	2a
Lazzarino (1989)	26	86	9 (35)	2b
Gisslinger (1991)	20	85	10 (50)	2c
Kasparu (1992)	14	86	0	2b
Berte (1996)	12	83	NA	2a/2b
Alvarado (2003)	11	100	2 (18)	Peg-2b
Saba (2005)	20	75	3 (15)	2a
Langer (2005)	36	75	13 (36)	Peg-2b
Samuelsson (2006)	21	70	11 (55)	Peg-2b
Jabbour (2007)	13	70	NA	Peg-2b
Quintás-Cardama (2009 and 2013)	39	81	NA	Peg-2a
Verger (2015)	31	100	39%	Peg-2b
Mascarenhas *et al.* (2016)	31	80	NR	Peg-2a
Gowin (2017)	20	65	NA	Peg-2a

ET: Essential thrombocythemia; IFN: Interferon; MPN: Myeloproliferative neoplasms; NA: Not applicable; NR: Not reported.

center, single-arm trial of 7.5 years follow-up that included 51 adults with low- and high-risk PV [19,21,22]. Complete hematological response according to a modified European LeukemiaNet (ELN) criteria was achieved in 64.3% and partial response for 33.3% of patients [21,22]. The complete hematological response response rates based only on hematological parameters and absence of a thromboembolic event was 80% [19]. Pooled safety data including 178 patients with PV dosed every 2–4 weeks showed that the most common adverse events were elevations in liver enzymes (20%), leukopenia (20%), thrombocytopenia (19%), arthralgia (13%), fatigue (12%), myalgia (11%) and flu-like illness (11%) [19].

SURPASS-ET clinical trial

SURPASS-ET (ClinicalTrials.gov, NCT04285086) is a global, open-label, randomized, phase III study to assess pharmacokinetics and compare the efficacy, safety and tolerability of ropeginterferon alfa-2b (P1101) with ANA in patients with ET who had HU resistance or intolerance. The study is currently active and recruiting patients.

Background & rationale

History of interferons in MPNs

In the USA, the pioneering study by Silver in 1988 documented the clinical effectiveness of IFN in controlling blood counts as well as pruritus and other constitutional symptoms in MPN patients [23]. Several clinical trials have been performed subsequently, using different preparations of IFN which are summarized in Table 1 [24–50].

In the abovementioned studies, various forms of IFN were used, and the possibility each of these IFN preparations have different effects that may influence response rates, disease evolution and allelic burden reduction. However, despite the heterogeneous response criteria used for evaluation and a possibility that each of these IFN preparations might have different effects that could have influenced response rates and disease evolution, notable response rates were observed. The efficacy and safety of IFN in treating ET is supported by positive clinical outcomes and the historic use of this treatment for more than 30 years.

Support for dosing & rationale

In the development program for ropeginterferon alfa-2b in viral hepatitis, doses up to 450 μg administered every 2 weeks were well-tolerated in subjects with chronic hepatitis C genotype 1 and genotype 2 (in combination with daily ribavirin) and chronic hepatitis B [25–27]. Almost all subjects reported adverse events during treatment that were mostly mild or moderate. Grade 3 and 4 toxicities were uncommon and were managed effectively with dose reduction or discontinuation. No unexpected safety signals of ropeginterferon alfa-2b up to 450 μg were noted in these studies [25–27].

In the development program for ropeginterferon alfa-2b in PV, specifically in the PROUD-PV/CONTINUATION-PV study, the dropout rate was low, and there were no new safety concerns observed with subjects receiving up to 500 μg in the maintenance period while the drug was effectively controlling elevated hematologic parameters [20]. The median dose was 400 μg at month 60 in the treatment arm [28]. Furthermore, the PEGINVERA trial identified 540 μg as the maximum tolerated dose of ropeginterferon alfa-2b in PV, which was the highest dose evaluated and no dose limiting toxicity was observed [21].

Population PK and PD analysis simulated for ET subjects with baseline platelet counts exceeding 450×10^9/l and a mean baseline WBC count of 11.4×10^9/l predicted that the subjects would achieve over a 60% reduction in platelets and WBCs starting by week 24 using a 250–350–500 μg ropeginterferon alfa-2b dosing scheme [28]. In addition, subjects tend to experience side effects of IFN early after the initiation of treatment and tolerability improves upon subsequent doses [29,30]. Findings from clinical development studies with ropeginterferon alfa-2b confirmed the safety of the higher starting doses exceeding 300 μg, further justifying the selected dosing scheme 250–350–500 μg being utilized in the phase III study.

Ropeginterferon alfa-2b will be administered in the clinic every 2 weeks as a subcutaneous injection during the study visits. The initial dose of ropeginterferon alfa-2b will be 250 μg, followed by 350 μg at week 2 and a target optimal dose of 500 μg by week 4 and onwards.

Study target population & comparator

HU in combination with a daily low-dose aspirin is the recommended therapy for ET in patients needing cytoreductive therapy [2,6]. HU is not indicated for the treatment of ET; it is approved in the USA and Europe for the treatment of certain cancers alone or in combination with other agents [5152]. However, HU resistance or intolerance is common [7,8,10]. ANA is included in the National Comprehensive Cancer Network (NCCN) and ELN guidelines as another treatment option for subjects with high-risk ET, despite concerns with efficacy and challenges with tolerability [2,6,12]. ANA is the only regulatory authority approved drug for ET. Pipobroman and busulphan are also included in the guidelines as second-line options after failure of primary therapy with either HU or IFN-α [2,6]. However, due to the potential leukemogenicity of these agents, and the relatively long life expectancy for subjects with PV and ET, it is recommended that these agents are reserved for elderly subjects (≥80 years) or those with advanced disease, where the risk of thrombosis outweighs the risk of myelodysplastic syndromes/AML [2,6,31].

SURPASS-ET: study design & methods

The SURPASS-ET trial (NCT04285086) is a global, phase III, open-label, multicenter, randomized, active-controlled study to assess pharmacokinetics and compare the efficacy, safety and tolerability of ropeginterferon alfa-2b to ANA as second line therapy for ET. Please see Figure 2 below for an overview of the study schematic. Approximately 160 patients will be randomized and recruited from 61 study sites across nine regions (USA, Australia, Canada, Taiwan, Japan, Korea, Hong Kong, China and Singapore). Patient visits will be scheduled every 2 weeks for both treatment arms. All patients will receive low-dose aspirin (75–150 mg/day) during the 12-month treatment period, unless contraindicated. The total duration of the study is approximately 14 months, with a 12-month treatment period and a 28 day screening and follow-up period. This is an open-label study, but patients will be randomized to the study drug assigned through interactive response technology and stratified by baseline platelet count, baseline total symptom score (TSS) and patient ethnicity (Asian and non-Asian). This study will assess the efficacy and safety of ropeginterferon alfa-2b by comparing it with the only approved treatment option, ANA, for patients who are HU-resistant or intolerant.

This protocol was approved by the Institutional Review Board/Independent Ethics Committee (IRB/IEC). This study will be conducted in compliance with the protocol, International Committee on Harmonization (ICH) Good Clinical Practice regulations and applicable regulatory requirements. Investigators will obtain an IRB/IEC approved signed informed consent from all subjects prior to exposure to study procedures, including screening for eligibility.

Eligibility criteria

The key inclusion and exclusion criteria are listed in Table 2. Major eligibility criteria include male or female subjects ≥18 years old with a diagnosis of high-risk ET, documented HU intolerance or resistance according to the

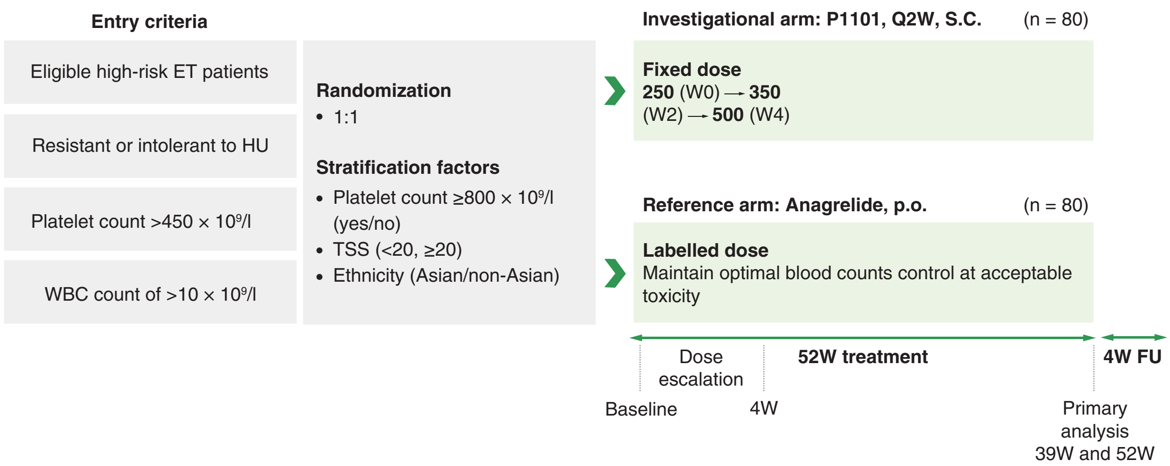

Figure 2. SURPASS-ET study schematic.
ET: Essential thrombocythemia; FU: Follow-up; HU: Hydroxyurea; l: Liter; p.o.: By mouth; S.C.: Subcutaneous; TSS: Total symptom score; W: Week; WBC: White blood cell.

modified ELN criteria [32] platelets $>450 \times 10^9$/l at screening, WBC $>10 \times 10^9$/l at screening and no history or presence of clinically relevant depression.

End points

The goal of treatment in patients with ET is to reduce thrombosis and hemorrhage risk, normalize symptoms, explore the clinical effects of allelic burden on driver mutations (*JAK2*V617F, *MPL*W515L/K and mutant *CALR*) and improve quality of life [2–5,33]. Studies have demonstrated that a higher allelic burden is associated with disease progression and poor survival, but further studies are needed to understand the therapeutic effects of reducing allelic burden in ET [33,34]. A key to achieve these treatment targets is to reduce platelets and, as shown by the recent data, WBCs, preferably to within the normal range [4,5].

The 2013 ELN and International Working Group-MPNs Research and Treatment (IWG-MRT) provides four response components for evaluation of response in ET [32]. Complete response requires: resolution of disease signs and improvement in symptoms ($\geq$10-point decrease in the MPN Symptom Assessment Form Total Symptom Score [MPN-SAF TSS] for at least 12 weeks; normalization of peripheral blood counts for at least 12 weeks; absence of vascular events and disease progression; and disappearance of bone marrow histological abnormalities).

The primary efficacy end point of this phase III trial is a durable response that utilizes the modified ELN response criteria [32] defined as meeting the following four categories: peripheral blood count remission (platelets $\leq 400 \times 10^9$/l and WBC $<9.5 \times 10^9$/l) and improvement or nonprogression in disease-related signs (splenomegaly) and large symptoms improvement or maintain nonprogression based on the MPN-SAF TSS and absence of hemorrhagic or thrombotic events and durability at both months 9 and 12. In order to assess improvement or nonprogression in ET-related splenomegaly, an ultrasound will be performed on all subjects with a palpable spleen at baseline. Improvement is defined as no palpable spleen at treatment visits and nonprogression is defined as an increased spleen size $\leq$25% as detected by ultrasound at treatment visits; progression is defined as an increased spleen size $>$25% on ultrasound at treatment visits. Progression is defined as the presence of a palpable spleen in subjects without a palpable spleen at baseline and an ultrasound must be performed. The changes from the MPN-SAF TSS scores from baseline will assess large symptom improvement or maintenance of nonprogression of ET. These changes are defined as the following: for baseline TSS scores $\geq$20, a 10-point score reduction; for baseline TSS scores inclusive of 15–19, a 5-point score reduction; for baseline TSS scores inclusive of 10–14, a score decrease to $\leq$10 points; and for a baseline TSS score $<$10, the score stays $<$10.

Secondary end points include durability of response at months 3 and 6, longitudinal rate of change in the ELN response rates over the 12 months, response rates based on peripheral blood count remission, no signs of progressive disease and absence of any hemorrhagic or thrombotic events at 3, 6, 9 and 12 months, occurrence of

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Table 2. Eligibility criteria for SURPASS-ET.

Key inclusion criteria	Key exclusion criteria
• Age ≥18 years • High-risk ET diagnosis[†] • Prior HU for ET[‡] • Interferon treatment-naive, or anti-P1101 binding antibody negative at screening[‡] • Documented resistance and/or intolerance to prior HU for ET[§] • Platelets >450 × 10^9/l at screening • WBC >10 × 10^9/l at screening • HGB ≥11 g/dl at screening for males and 10 g/dl at screening for females • Adequate hepatic function[¶] • Creatinine clearance ≥40 ml/min[#] • Males and females of childbearing potential, as well as all women <2 years after the onset of menopause, must agree to use an acceptable form of birth control until 28 days following the last dose of the study drug and females must agree to not breastfeed during the study • Written informed consent obtained from the subject and ability for the subject to comply with the requirements of the study	• Any contraindications or hypersensitivity to IFN-alfa or ANA and their excipients • Known risk factors for QT-prolongation[††] • Co-morbidity with severe or serious condition that, in the Investigator's opinion, would jeopardize the safety of the subject or their compliance with the protocol, including significant cardiac disease (including NYHA Class III–IV congestive heart failure and clinically significant arrhythmias) and pulmonary hypertension • History of major organ transplantation • Pregnant or lactating females • Use of any investigational drug <4 weeks prior to the first dose of study drug or not recovered from effects of prior administration of any investigational agent • Subjects with documented ANA resistance or intolerance[‡‡] • Subjects with any other significant medical conditions that, in the opinion of the Investigator, would compromise the results of the study or may impair compliance with the requirements of the protocol, including but not limited to: ○ Documented autoimmune disease at screening or in the history[§§] ○ Clinically relevant pulmonary infiltrates, pneumonia, and pneumonitis at screening that, in the Investigator's opinion, would jeopardize the safety of the subject or their compliance with the protocol ○ Infections with systemic manifestations[¶¶] ○ Evidence of severe retinopathy[##] or clinically relevant ophthalmological disorder[†††] ○ History or presence of clinically relevant depression ○ Previous suicide attempts or at any risk of suicide at screening, in the judgement of the Investigator ○ History or presence of clinically significant neurologic diseases ○ History of any malignancy within 5 years[‡‡‡] ○ History of alcohol or drug abuse within the last year ○ History or evidence of any other MPN ○ History of splenectomy

[†]According to the WHO 2016 Criteria (either >60 years and *JAK2*V617F-positive at screening, or having disease-related thrombosis or hemorrhage in the past).
[‡]The washout period between the last dose of HU or interferon and randomization should not be shorter than 7 days.
[§]According to the modified ELN criteria, whereby at least one of the following criteria is met:
- Platelet count >600 × 10^9/l at ≥2 g/day (or ≥2.5 g/day if subject body weight >80 kg) or maximally tolerated dose if <2 g/day after at least 3 months of HU;
- Platelet count >400 × 10^9/l and WBC count <2.5 × 10^9/l at any dose and duration of HU;
- Platelet count >400 × 10^9/l and HGB <10 g/dl at any dose and duration of HU;
- Presence of HU-related toxicities at any dose and duration of therapy (e.g., leg ulcers, mucocutaneous manifestations, pneumonitis or HU-related fever).

[¶]Defined as bilirubin ≤1.5 × ULN, PT (INR) ≤1.5 × ULN, albumin >3.5 g/dl, ALT ≤2.0 × ULN, AST ≤2.0 × ULN at screening.
[#]According to the Cockcroft–Gault equation.
[††]Examples of QT-prolongation include congenital long QT, known history of acquired QT-prolongations.
[‡‡]The items below need to be excluded for all ANA pre treated patients at baseline or in the documented medical history.
- Resistance: platelets >600 × 10^9/l at ANA dose of at least 3 mg/day or at the subject's highest individual tolerable dose after treatment duration of at least 8 weeks.
- Intolerance (at least one of the following criteria is met):
  1. Clinically significant arrhythmia (including Torsades de Pointes) on the treatment with ANA;
  2. Suspected or verified pulmonary toxicity (interstitial lung diseases) caused by the drug (any grade or severity);
  3. Suspected or verified pulmonary hypertension caused by the drug (any grade or severity);
  4. Episodes of significant bleeding with HGB decrease of >1 g/dl;
  5. Any drug-related toxicity of grade 3 or higher, which re-appeared after repeated therapy with ANA, while the duration of each episode was longer than 72 h and it required sustained dose reduction of ANA;
  6. Any drug-related toxicity of ANA, which did not allow re-initiation of the therapy due to medical reasons in the opinion of the treating physician in reference to the ANA label.

[§§]Examples include thyroid dysfunction, hepatitis, idiopathic thrombocytopenic purpura, scleroderma, psoriasis or any arthritis of autoimmune origin.
[¶¶]Examples include bacterial, fungal or HIV, except HBV and/or HCV, at screening.
[##]Examples include cytomegalovirus retinitis and macular degeneration.
[†††]Due to diabetes mellitus or hypertension.
[‡‡‡]Except stage 0 CLL, basal cell, squamous cell and superficial melanoma.
ANA: Anagrelide; CLL: Chronic lymphocytic leukemia; ELN: European LeukemiaNet; ET: Essential Thrombocythemia; HGB: Hemoglobin; HBV: Hepatitis B virus; HCV: Hepatitis C virus; HU: Hydroxyurea; INR: International normalized ratio; MPN: Myeloproliferative neoplasm; NYHA: New York Heart Association; PT: Prothrombin time; ULN: Upper limit of normal; WBC: White blood cell.

thromboembolic events, time to first peripheral blood count remission response, duration of peripheral blood count remission response, symptomatic improvement assessed by the EuroQoL 5 dimensions 3 level version questionnaire (EQ-5D-3L), symptomatic improvement assessed by the 10-item MPN-SAF TSS (MPN10) and change of *CALR*, *MPL* or *JAK2*V617F allelic burden from baseline over time [32].

Exploratory end points include an optional bone marrow histological remission defined as the disappearance of megakaryocyte hyperplasia and absence of >grade 1 reticulin fibrosis.

Adverse events are graded according to Common Terminology Criteria for Adverse Events (CTCAE; version 5.0) and captured throughout the study. Adverse events of special interest included cardiovascular events, hemorrhagic, thrombotic events and psychiatric events.

Pharmacokinetic parameters of ropeginterferon alfa-2b, including (but not limited to) C_{min}, T_{max}, C_{max}, and $AUC_{0\text{-}t}$, will be derived using PPK analysis and the relationship between exposure and efficacy and safety end points will be examined using E–R analysis. In addition, the proportion of subjects who are anti-drug antibody positive as well subjects with as neutralizing antibody in ropeginterferon alfa-2b arm will be determined.

Statistical analysis methods

The statistical analysis will be done in agreement with the International Conference on Harmonization (ICH) Topics E3, E6 and E9 (including E9R1), Statistical Principles for Clinical Trials and Statistical Analysis Plan finalized and approved before database lock. Analyses will be conducted using Statistical Analysis System (SAS®) software, Version 9.4 or higher (SAS/STAT 12.1, SAS Institute, NC, USA). Categorical and continuous variables will be summarized by treatment and visits, as appropriate. PK parameters will be calculated using population PK analysis or noncompartmental methods, as appropriate. The relationship between exposure and efficacy and safety end points will be determined using exposure–response analysis. Cochran–Mantel–Haenszel test will be used for the treatment comparison on the primary end point.

Interim analysis

Interim analysis may be implemented for a possible sample size adjustment according to the data safety and monitoring board (DSMB) recommendation. If the incidence rates of hemorrhagic and thrombotic events are less than 20% across both treatment arms, then no sample size re-estimation will be needed. If the incidence rates for these events crosses the prespecified threshold, the DSMB will be asked to examine the distribution of incidence rates between treatment arms and the sponsor will remain blinded. If reassessing the sample size is deemed necessary by the DSMB, then a sample size re-estimation is planned when approximately 70 subjects have completed their 12 month visit and primary end point data are available. If conducted, the sample size may be recalculated by a DSMB statistician using the conditional power approach based on the interim data. If needed, the sample size will be adjusted upward only when the conditional power of reaching significance at the final analysis with the original sample size (130 randomized subjects) is greater than 50%. No early stopping rule for claiming efficacy is planned. The type I error rate will not be inflated; thus no adjustment is needed.

Conclusion

Ideal therapeutic approaches for patients with high-risk ET should achieve adequate cytoreduction, reduce the risk of thrombo-hemorrhagic complications, and prevent progression to post-ET-MF or secondary AML. There is a strong rationale and evidence for using IFNs in the treatment of ET. Ropeginterferon alfa-2b is an interferon with a unique PK/PD parameters and an acceptable tolerability, and safety profile. Ropeginterferon alfa-2b (P1101) will be evaluated as a second-line agent compared with ANA for the treatment of ET in the global SURPASS-ET trial. This represents the first time that a pegylated IFN will be evaluated in a head-to-head trial with an approved ET treatment. The results will shed light on the role of ropeginterferon alfa-2b as a second-line agent in the treatment of high-risk ET as well as its impact on disease modification as assessed by the allelic burden.

Financial & competing interests disclosure

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USA Corp: Current Employment. W Shih: PharmaEssentia Corporation: Consultancy. O Zagrijtschuk: PharmaEssentia U.S.A. Corp.: Current Employment. C Zimmerman: PharmaEssentia U.S.A. Corp. S Verstovsek: Gilead: Research Funding; Protagonist Therapeutics: Research Funding; NS Pharma: Research Funding; Incyte Corporation: Consultancy, Research Funding; PharmaEssentia: Research Funding; Ital Pharma: Research Funding; CTI BioPharma: Research Funding; Blueprint Medicines Corp: Research Funding; AstraZeneca: Research Funding; Promedior: Research Funding; Genentech: Research Funding; Celgene: Consultancy, Research Funding; Roche: Research Funding; Novartis: Consultancy, Research Funding; Sierra Oncology: Consultancy, Research Funding; Constellation: Consultancy; Pragmatist: Consultancy. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

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Executive summary

Essential thrombocythemia

- Hydroxyurea is a nondisease-specific cytoreductive therapy utilized in high-risk essential thrombocythemia (ET) but it is associated with high rates of intolerance and resistance.
- Anagrelide is an approved second-line treatment for ET after hydroxyurea failure, but it is associated with toxicities and disease progression.
- Interferon alfa has been studied in myeloproliferative neoplasms for over 30 years. Ropeginterferon alfa-2b has improved tolerability and pharmacokinetics compared with earlier interferons making it an appealing treatment option for ET.

Ropeginterferon alfa-2b in myeloproliferative neoplasms

- Ropeginterferon alfa-2b is approved for the treatment of adults with polycythemia vera in the USA and Europe based on safety and efficacy data from the PEGINVERA and PROUD-PV/CONTINUATION-PV studies.
- This is the first time ropeginterferon alfa-2b will be evaluated in patients with high-risk ET.

SURPASS-ET

Study design

- This is a phase III, open-label, multicenter, randomized, active-controlled study to assess pharmacokinetics and to compare the efficacy, safety and tolerability of ropeginterferon alfa-2b (P1101) compared with anagrelide as second-line therapy after 12 months of treatment for patients with ET who have shown resistance or intolerance to hydroxyurea.
- P1101 will be administered subcutaneously every 2 weeks at the starting dose of 250 μg, followed by 350 μg at week 2 and target optimal dose of 500 μg at week 4 and fixed throughout the treatment period, unless dose adjustment is needed for safety or tolerability reasons.
- The treatment period will be 12 months with visits scheduled every 2 weeks in both treatment arms and a safety follow-up visit will take place 28 days after the end of treatment.

Eligibility criteria

- Key eligibility criteria include adults with high-risk ET who have documented resistance or intolerance with prior use of hydroxyurea, platelets $>450 \times 10^9$/l at screening, WBC $>10 \times 10^9$/l at screening, and without a history of clinically relevant depression.

Outcome measures

- The primary efficacy end point is the assessment of the response rate utilizing the modified European LeukemiaNet criteria.
- Key secondary efficacy end points include durability of a response at months 3 and 6, longitudinal rate of change in the European LeukemiaNet response over the treatment period of 12 months, and occurrence of thromboembolic events.

Conclusion

- Ropeginterferon alfa-2b is a site-specific monopegylated interferon alfa-2b with unique PK:PD parameters and an acceptable safety and tolerability profile. The SURPASS-ET trial is the first time that a pegylated interferon is being evaluated in a head-to-head trial with an approved ET treatment.

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